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Survey Paper

A survey of graph neural networks and their industrial applications

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ABSTRACT

Graph Neural Networks (GNNs) have emerged as a powerful tool for analyzing and modeling graph-structured data. In recent years, GNNs have gained significant attention in various domains. This review paper aims to provide an overview of the state-of-the-art graph neural network techniques and their industrial applications. First, we introduce the fundamental concepts and architectures of GNNs, highlighting their ability to capture complex relationships and dependencies in graph data. We then delve into the variants and evolution of graphs, including directed graphs, heterogeneous graphs, dynamic graphs, and hypergraphs. Next, we discuss the interpretability of GNN, and GNN theory including graph augmentation, expressivity, and over-smoothing. Finally, we introduce the specific use cases of GNNs in industrial settings, including finance, biology, knowledge graphs, recommendation systems, Internet of Things (IoT), and knowledge distillation. This review paper highlights the immense potential of GNNs in solving real-world problems, while also addressing the challenges and opportunities for further advancement in this field.

1. Introduction

In the realm of machine learning and artificial intelligence, the recent surge in the efficacy of neural networks has heralded a new era in pattern recognition and data mining. This transformation has been particularly pronounced in the domain of Graph Neural Networks (GNNs), a sophisticated paradigm that has gained traction for its adeptness in handling graph-structured data. Graphs, as a fundamental data structure, encapsulate a set of objects (nodes) and their interconnections (edges), finding ubiquitous applications ranging from social networks [1] and physical systems to intricate knowledge graphs [2] and biological networks [3]. The versatility and pervasiveness of graphs across numerous fields underscore their significance and the necessity for advanced modeling and analysis like GNNs.

At the heart of GNNs' rising prominence is their unique ability to manage the complexities inherent in graph data. Unlike traditional Euclidean data (e.g., images, text, and videos), graph data often exhibit non-regularity and variability in node connectivity and structure. Drawing inspiration from the successes of Convolutional Neural Networks (CNNs) [4] and Recurrent Neural Networks (RNNs) [5,6] in processing Euclidean data, GNNs extend these concepts to accommodate the irregularities of graph data. A seminal aspect of this adaptation is the generalization of operations like convolution from 2D Euclidean space to graph domains. For instance, an image can be considered as a special graph where pixels are nodes and edges only connect adjacent pixels.

Furthermore, GNNs address the limitations of earlier graph processing methods by integrating advanced deep learning techniques [7]. Early approaches to neural networks for graphs in the 1990s, such as Recursive Neural Networks [8,9], laid the groundwork, but they were constrained in their representational capacity and scalability. The advent of deep learning, especially with CNNs, reinvigorated interest in graph-based neural networks, leading to the development of more robust and versatile GNN architectures. These architectures effectively leverage multi-scale and localized spatial features of graphs, resulting in highly expressive representations that are vital for diverse applications.

Another driving force behind the evolution of GNNs is the industrial need for outstanding graph representation learning. Such learning aims to encode graph nodes, edges, or subgraphs into low-dimensional vectors, mirroring the success of word embeddings in natural language processing. Early graph embedding methods like DeepWalk [18] and node2vec [19] showcased the potential of representation learning for graphs, but they also highlighted challenges such as computational inefficiency and lack of generalization to dynamic or unseen graphs. GNNs overcome these hurdles by enabling parameter sharing across

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Table 1
Surveys on theory and applications of graph neural networks.

Field	Paper	Context	Year
General GNN	Zhou et al. [10]	Graph mining, Physics, Chemistry, Biology, Knowledge graph, Generation, Combinatorial optimization, Traffic network, Recommendation systems, Text, Image	2020
	Wu et al. [11]	Computer vision, Natural language processing, Traffic, Recommender systems, Chemistry	2020
	Waikhom et al. [12]	Natural language processing, Computer vision, Image classification, Social networks, E-Commerce, Recommender systems, Chemistry, Drug discovery, Physical networks, Traffic, Circuit design	2022
Adversarial GNN	Chen et al. [13]	Taxonomies of GNN attack strategy and defense methods	2020
Heterogeneous GNN	Dong et al. [14]	Systematic categorization and analysis of the merits of various existing heterogeneous network embedding algorithms	2020
	Zheng et al. [15]	Systematic Taxonomy of heterophilic GNNs	2022
Dynamic GNN	Barros et al. [16]	A novel taxonomy for dynamic graph embedding input and output	2021
Hypergraph GNN	Antelmi et al. [17]	A new taxonomy of hypergraph embedding methods	2023

nodes, thus exhibiting superior generalization capabilities and making them adept at handling dynamic and heterogeneous graph structures.

There exist several comprehensive reviews on graph neural networks. A detailed review over existing graph neural network models and their variants with categorized application scenarios are presented in [10]. Wu et al. [11] propose a new taxonomy of GNNs into four groups: recurrent GNNs, convolutional GNNs, graph autoencoders, and spatial–temporal GNNs. It provides detailed descriptions of representative models for each type of GNN. Theoretical and empirical aspects of GNNs and various applications with popular graph-based datasets are introduced in [12]. Work [12] focuses more on the taxonomy of GNN models and detailed convolution and pooling operators defined on graphs.

Several survey papers focus on more specific fields of GNNs. Chen et al. [13] give detailed overviews of adversarial learning methods on graphs in both graph data attack and defense. Dong et al. [14] and Zheng et al. [15] provide a review of heterogeneous network representation learning. Barros et al. [16] focus on graph embedding on dynamic graphs and discuss different dynamic behaviors in dynamic networks. Antelmi et al. [17] elaborate on existing literature on hypergraph representation learning problems and methods. A summary of above survey is given in Table 1.

In this paper, we introduce preliminary knowledge of graphs and their variants to help readers acquire the required understanding of the given domain. In addition to that, we offer an extensive analysis of the current state of research on GNNs, showcasing their considerable potential in tackling real-world challenges. We also identify key obstacles and opportunities for advancing the GNN field further.

The rest of this survey paper is organized as follows. Section 2 introduces the basic elements of GNN and commonly used GNN structures. Section 3 introduced the variants and evolution of graph and their related algorithms. In Section 4 we provide an overview of the interpret methods in GNN to help enhance the understanding on how GNNs work in industrial applications. Section 5 provides some of the theoretical studies related to industrial application. Section 6 shows a clear application-oriented taxonomy. We discuss the evaluation and dataset in Section 7. In Section 8, we provide some directions on the challenges in GNNs. The paper is concluded in Section 9.

2. Fundamentals of GNNs

The concept of graph neural networks was first formally introduced by Franco et al. [20], who also integrated the idea of recurrence into graph networks based on the fixed-point theorem. Despite the initial success of these approaches, they were primarily built upon the notion of state transition systems on graphs, where iterations continued until convergence. This methodology, however, inherently limited the model's extendability and representational capacity. A significant advancement occurred when Bruna et al. [21] introduced the spatial and spectral construction of graphs, which allowed the application of convolutional neural networks in the graph domain, marking a pivotal shift in the field. Building on this foundation, Kipf et al. [22] proposed the variational graph autoencoder, an extension of their earlier work on graph convolutional networks [23]. This was followed by the introduction of the attention mechanism into GNNs, resulting in the Graph Attention Network model, further enhancing the model's ability to focus on relevant parts of the graph during processing.

Table 2
Summary of abbreviations.

Abbreviation	Full form
GNNs	Graph Neural Networks
CNNs	Convolutional Neural Networks
RNNs	Recurrent Neural Networks
GRNNs	Graph Recurrent Neural Networks
GCNs	Graph convolutional networks
GAEs	Graph Autoencoders
GGNN	Gated Graph Neural Network
HGNN	Hypergraph Neural Networks
LSTM	Long Short Term Memory
SSL	Self-Supervised Learning
KG	Knowledge Graph
IoT	Internet of Things
KD	Knowledge Distillation
QoS	Quality-of-Service

We have summarized the abbreviations used in our paper in Table 2.

2.1. Basic elements of GNN

A graph $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$ is a data structure defining a set of nodes denoted as V and their relationships denoted as E. Here are the basic elements of a graph data.

- **Node (Vertex):** Node is the fundamental unit in a graph, typically denoted as $v_i \in \boldsymbol{V}$. The people are the nodes in a social network and their related information is the nodes features denoted as $x_i \in \boldsymbol{X}$.
- **Edge:** The relationship between nodes, usually represented as $e_{ij} = (v_i, v_j) \in \boldsymbol{E}$. Based on the relationship between nodes, the edges can be directed or undirected, homogeneous or heterogeneous.
- **Neighbors:** The neighbors of node v_i, denoted as $N(v_i)$, is the set of nodes directly connected to v_i via edges.
- **Adjacency Matrix:** An $N \times N$ matrix $\boldsymbol{A}$, where $\boldsymbol{A}_{ij}$ indicates the presence of an edge between nodes v_i and v_j. If an edge exists, the element is 1 or the weight of the edge, otherwise 0.

- **Feature Matrix:** An $N \times K$ matrix $\boldsymbol{X}$, where each row x_i represents the feature vector of node v_i.

The basic idea of graph neural networks is to learn representations of nodes and edges in graph structures by propagating information and updating features through nodes. The main processes include:

> **Message Passing:** In each layer, a node receives information from its neighbors and aggregates it through a certain mechanism. The goal of this step is to update its representation using the information from its neighbors.
> **Node Feature Update:** Each node updates its feature representation using the aggregated neighbor information and its own features, usually achieved through a nonlinear transformation (such as a neural network layer).
> **Readout Layer:** After multiple rounds of message passing and feature updates, the final node representations can be used for node-level tasks (e.g., node classification) or aggregated into graph-level representations for graph-level tasks (e.g., graph classification).

2.1.1. Message passing and aggregation

Message passing is the core mechanism of GNNs. In each layer, a node v collects information from its neighboring nodes $\boldsymbol{N}(v)$ and aggregates it. The aggregation can be a simple sum, average, or a more complex function.

- **Message Passing Formula:**

  $$m_v^{(k)} = \text{AGGREGATE}(\{x_u^{(k-1)} \mid u \in \boldsymbol{N}(v)\})$$

  Here, $m_v^{(k)}$ is the aggregated information for node v at layer k, and AGGREGATE is the aggregation function.
- **Common Aggregation Functions:**
  - **Sum:** Directly sums up the features of neighboring nodes.

    $$m_v^{(k)} = \sum_{u \in \boldsymbol{N}(v)} x_u^{(k-1)}$$

  - **Mean:** Computes the average of the features of neighboring nodes.

    $$m_v^{(k)} = \frac{1}{|\boldsymbol{N}(v)|} \sum_{u \in N(v)} x_u^{(k-1)}$$

  - **Max:** Takes the maximum value from the features of neighboring nodes.

    $$m_v^{(k)} = \max_{u \in \boldsymbol{N}(v)} x_u^{(k-1)}$$

2.2. Graph Recurrent Neural Networks (GRNNs)

Graph recurrent neural networks (GRNNs) are foundational to the development of GNNs technology. Their primary objective is to generate representations of nodes using recurrent neural network architectures like Long Short Term Memory (LSTM) [24] or Gated Graph Neural Network (GGNN) [25]. This involves a process in Fig. 1 where nodes repetitively share information or messages with their neighboring nodes using local transition function f_ω as:

$$x_i(t+1) = f_\omega(l_i, l_{NE}, x_{NN}(t), l_{NN}), \tag{1}$$

where $x_i(t)$ is the represented vector of node i at the tth iteration, l_i is the label of node i, l_{NE} is the set of labels from the edges of node i, x_{NN} is the set of represented vectors from the neighbor of node i and l_{NN} is the set of labels of neighbor nodes of node i.

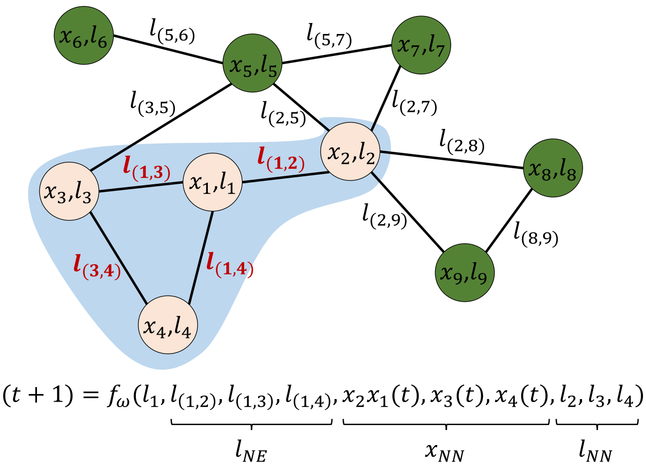

Fig. 1. GRNN aggregation process.

2.3. Graph Convolutional Networks (GCNs)

Graph convolutional networks (GCNs) extend the concept of convolution, traditionally applied to structured grid data, to the domain of graph data. The fundamental principle behind GCNs involves the aggregation of a node's features with those of its neighbors to construct a comprehensive representation of the node. Specifically, for a given node v, its representation is derived by integrating its own features, denoted as x_v, with the features of its neighboring nodes, denoted as x_u, where u belongs to the set of v's neighbors $N(v)$. This approach is distinct from GRNNs, as GCNs employ multiple layers of graph convolution to progressively extract higher-level features of nodes, rather than relying on recurrent mechanisms. GCNs are foundational in the development of numerous advanced GNN models, serving as a critical component in the architecture of complex GNN systems. The general structure is illustrated in Fig. 2.

For instance, to deal with Quality-of-Service (QoS) data, a Two-stream Graph convolutional network-incorporated Latent Feature Analysis has been proposed in [26]. A biparty graph is designed in the model to represent the user-service interactions, and a GCN is used to learn the high-order connectivity in QoS data. NODE-SELECT [27] uses subset layers which only allow the best sharing-fitting nodes to propagate their information in order to effectively target noise propagation during the message-passing procedure. They estimate a random selectivity value $\hat{p}_i$ as a measure to classify nodes with potentially harmful or useless embedding information. With a determined threshold T, the selection of a neighbor node is performed via:

$$S(v_i) = \begin{cases} 1, & \hat{p}_i \geqslant T \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $\hat{p}_i = \sigma\left(\mathbf{W}_0\left(\sum_{j\in\mathcal{N}_{(i)}} \mathbf{W}x_j\right)\right)$, and $\mathbf{W}_0 \in \mathbb{R}^{1\times F'}$ is a weight matrix. Then, the transformed features only aggregate on the selected neighbors.

Facing the limitations of over-smoothing, non-robustness, and weak-generalization in semi-supervised learning on graphs, Graph Random Neural Networks (GRAND) was proposed by Feng et al. [28]. GRAND decouples feature propagation and transformation to avoid over-smoothing. It uses a random propagation strategy for graph data augmentation and leverages consistency regularization to optimize the prediction consistency of unlabeled nodes across different data augmentations. GRAND+ [29] overcomes the limitation of GRAND in large-scale graph computation since its effectiveness relies on computationally expensive data augmentation procedures. The work [29] develops a generalized forward push algorithm to pre-compute a general propagation matrix and employ it to perform graph data augmentation

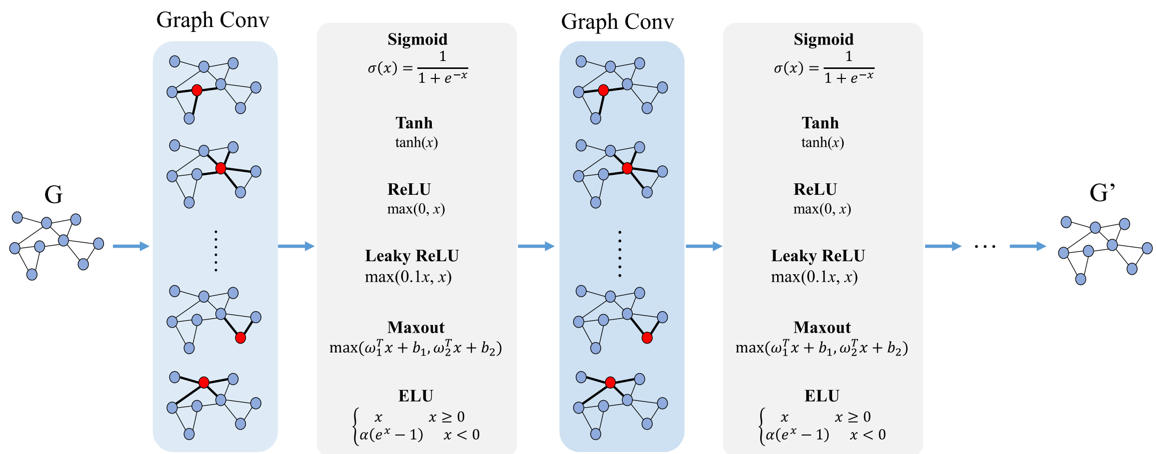

Fig. 2. GCN general structure.

in a mini-batch manner. ωGNN [30] incorporates trainable channel-wise weighting factors ω to learn and mix multiple smoothing and sharpening propagation operators at each layer to increase GNN's expressiveness and avoid the over-smoothing problem. LazyGNN [31] focuses on capturing long-distance dependency in graphs by shallower models instead of deeper models to be more scalable on large-scale graphs. LazyGNN leverages the computation redundancy between training iterations and reuses the propagated information as training proceeds. DAGNN [32] aims to effectively incorporate information from large receptive fields while avoiding the common pitfalls of existing deep GNN models. DAGNN utilizes a single propagation layer that incorporates different levels of neighbor representation after transformation based on generated retainment scores. DAGNN ensures that each node in the graph can effectively integrate and learn from its neighborhood information while maintaining distinct node features. Another recent work by Chen et al. [33] verifies that smoothing is the nature of most typical graph convolutions. It is shown that reasonable smoothing makes graph convolutions work; while over-smoothing results in poor performance.

2.4. Graph Autoencoders (GAEs)

Graph Autoencoders (GAEs) are unsupervised learning frameworks designed to encode nodes or entire graphs into a latent vector space and then reconstruct graph data from this encoded information. They are commonly used for learning network embeddings, graph generative distributions, or structure prediction [34]. For network embedding, GAEs learn latent node embedding Z by reconstructing the structural information of a graph such as an adjacency matrix. Then graph G' with the latent embedding Z is regenerated. In terms of graph generation, some methods incrementally generate nodes and edges, while others output an entire graph in one step. The work [35] proposes a novel graph decoder to reconstruct the entire neighborhood information regarding both proximity and structure via Neighborhood Wasserstein Reconstruction. It describes decoding as iteratively sampling from probability distributions over multi-hop neighbors' representations obtained through the GNN encoder. The decoder takes node degrees, neighbor-representation distributions, and node features information into consideration and adopts an optimal-transport loss based on Wasserstein distance. GraphMAE [36] is a masked graph autoencoder that focuses on node feature reconstruction. It re-masks the encoder's output embeddings of masked nodes before they are fed into the decoder and employs cosine error during the reconstruction. Hua et al. [37] incorporate a GAN structure into GAE to enhance semantic-relevant feature representation during encoding and address near-convergence in GAN-based AEs.

2.5. Attention-based GNN

Attention mechanism [38,39] is inspired by the human visual system, which imitates the human cognitive awareness about specific information to focus more on the critical aspects of data. The attention mechanism in GNNs allows the neural networks to learn a dynamic and adaptive aggregation of the neighborhood while filtering out noise in the graph. Attention-based GNNs can be categorized into three types: local attention, global attention, and feature fusion attention, depending on the underlying attention functions.

GAT [40], known as the graph attention network, is the pioneer in local attention GNNs. GAT only computes the masked attention for each node with its first-order neighbors and assigns different weights to nodes in the feature aggregation.

To eliminate noisy high-frequency information from the graph, MAGNA [41] captures large-scale structural information in each layer with a low-pass filter. The model propagates the attention scores across the graph with Personalized PageRank, increasing the receptive field for each layer.

Hyperbolic attention GNNs [42] introduce a dual attention mechanism that simultaneously captures structural properties and feature similarities between nodes in hyperbolic space, enhancing the representation of hierarchical graph structures.

Dual separated attention GNN [43] leverages an attention-based label propagation mechanism to refine label predictions through adaptive neighbor aggregation, thereby enhancing classification robustness in label-deficient settings.

A Multimodal Graph Attention Network [44] is proposed to leverage GNNs with gated attention mechanisms to effectively capture and disentangle user preferences across different modalities. The model improves upon prior models by integrating attention mechanisms to control information flow and enhance prediction accuracy.

3. Variants and evolution of graphs

Real-world graph data usually have a more complex structure than an undirected graph. The problem settings are determined based on the structure and size of the graph type found in data and we have a simple illustration of these graph types in Fig. 3.

3.1. Directed graphs

A directed graph $\boldsymbol{G}_{Dir} = (\boldsymbol{V}, \boldsymbol{E}_{Dir})$ usually contains more information than undirected graph. It contains a set of nodes as $\boldsymbol{V}$ and a set of ordered pairs of edges as $\boldsymbol{E}_{Dir}$. Adding edge direction provides substantial information about the relationships among the nodes in a non-asymmetric way. For example in a finance network of money flow,

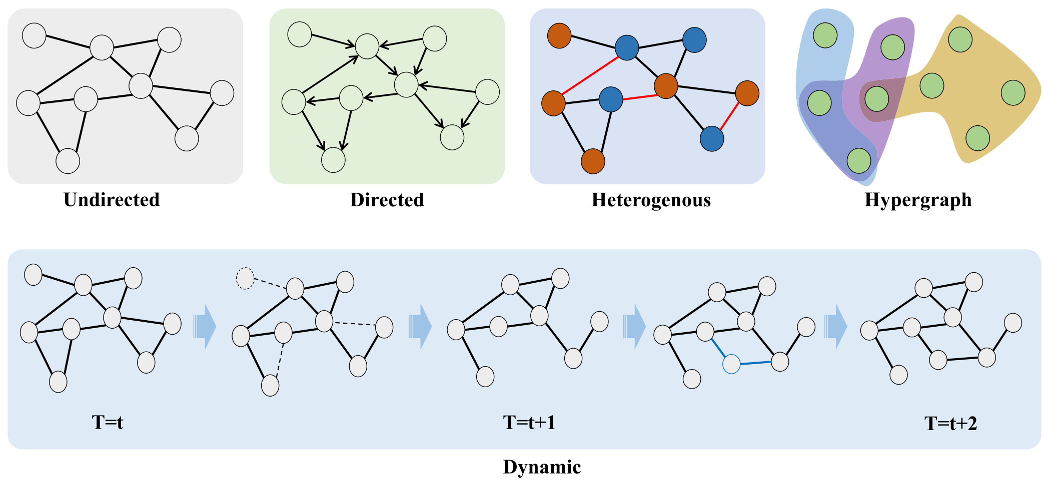

Fig. 3. Example of variants of graphs.

we can use directed edges to represent the direction of money flow between buyer and seller and turn it into a directed graph. DGP [45] uses descendant and ancestor propagation to combine information of different directions. Complex Hermitian matrices such as the magnetic Laplacian are used to help relate geometric characteristics of directed graphs to properties of eigenvectors and eigenvalues of graph Laplacians and related matrices in spectral graph theory [46].

3.2. Heterogeneous graph

A heterogeneous graph $\boldsymbol{G}_H = (\boldsymbol{V}, \boldsymbol{E}, \phi, \psi)$ consists of multiple node types $\phi(\boldsymbol{V}) = \{\phi_1, \ldots, \phi_m\}$ or edge types $\psi(\boldsymbol{E}) = \{\psi_1, \ldots, \psi_m\}$. Each node $v_i \in \boldsymbol{V}$ is associated with a node type $\phi(v_i)$ and each edge $e_{ij} \in \boldsymbol{E}$ is associated with a edge type $\psi(e_{ij})$. An easy example is the co-author and paper-citation networks. They have node types of authors, papers and venues and edge types of citation and authorship.

To consider high-order local structures and node homophily, a Semantics and Homophily preserving Network Embedding (SHNE) model [47] is proposed, which uses motifs to capture higher order connectivity patterns for structural semantics and together with feature similarity to discover potentially correlated nonlocal neighbors.

To adaptively learn graph structures for HGNNs, the work [48] introduces graph structure learning into heterogeneous graphs. It learns a heterogeneous graph and GNN parameters jointly based on the fused heterogeneous graph by using feature similarity, feature propagation information, and semantic information from each relation subgraph. The work [49] combines different singlelayer long metapaths structure and transformer-based semantic fusion module to better utilize semantic information.

3.3. Dynamic graph

A dynamic graph $\boldsymbol{G}_{dyn} = (\boldsymbol{V}, \boldsymbol{E}, \boldsymbol{T})$ is a type of graph that changes over time. $\boldsymbol{V} = V(t)$ is a collection of node sets at time $t \in \boldsymbol{T}$ and $\boldsymbol{E} = E(t)$ is a collection of edge sets at time $t \in \boldsymbol{T}$. For every $t \in \boldsymbol{T}$, we can have a graph snapshot $G(t) = (V(t), E(t))$ representing the dynamic graph at timestamp t. Nodes are allowed to add or delete, and the corresponding edges between the nodes can also be created or updated.

Zhou et al. [50] empirically and theoretically elucidated the existence of topology-task discordance on dynamic graphs. They proposed GReTo with intra-graph and inter-graph homophily measures to capture diverse spatial–temporal relations and devised an adaptive layer-wise importance measurement to promote the immediate neighborhood aggregation towards high-order propagation.

Structural insights are incorporated at both local and global levels with frequent used first-order temporal neighbor information to generate more informative node representations [51]. Liu et al. [52] focuses on the representation learning in unsupervised scenarios. They introduces deep clustering techniques to suit the interaction sequence-based batch-processing pattern of temporal graphs. Luo et al. [53] focuses on interacting behaviors prediction between multiple nodes from their joint neighborhood. They use dictionary-type neighborhood representations and a GPU-based neighborhood cache system to enabling parallel processing of structure features and fast link predictions. Temporal Multi-Modal Graph Learning is studied by constructing temporal graphs for acoustic events, where audio and video segments are nodes, and temporal relationships are edges, capturing dynamic intra- and inter-modal information [54]. The Hawkes process is introduced to enhance modeling of these interactions. A Dynamic Graph Information Bottleneck framework is introduced to learn robust and discriminative representations against adversarial attacks [55]. They put forward the Minimal-Sufficient-Consensual Condition to identify optimal representations that are both informative and robust for prediction in dynamic environments. A novel defense framework is presented to redefine backdoor attack prevention in federated learning as a dynamic attributed graph clustering problem [56]. It accurately detects malicious clients, ensures convergence, and significantly reducing attack success rates across various datasets.

3.4. Hypergraph

A hypergraph is an ordered pair $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$, where $\boldsymbol{V}$ is the set of nodes, and $\boldsymbol{E}$ is the set of hyperedges. Each hyperedge is a non-empty subset of nodes. Instead of the traditional adjacent matrix, a hypergraph is usually represented by an incidence matrix $\boldsymbol{H} \in \{0,1\}^{|V|,|E|}$. $H_{v,e}$ indicating whether vertex v is in hyperedge e.

General Hypergraph Neural Networks (HGNN+) [57] is a general hypergraph neural network framework to handle the challenges on representation learning using high-order correlations, as the extension version of Hypergraph Neural Networks (HGNN) [58].

Contrastive learning is leveraged to capture spatial correlations and utilizes the Hawkes process for dynamic interaction within specific historical scopes, effectively addressing both spatial and temporal correlations in prediction [59]. Hypergraph Collaborative Network [60] consideration information from both previous vertices and hyperedges to achieve informative latent representations by alternately updating them and incorporate hypergraph reconstruction error as a regularizer to further improve the effectiveness of the learned classifier.

4. Interpretability

GNNs are increasingly recognized for their proficiency in handling graph-structured data. Despite their successful applications in complex tasks like recommendation systems, traffic forecasting, and medical diagnosis, GNNs face a critical challenge: their internal workings are akin to a "black box", making it difficult for humans to understand the rationale behind their predictions. In decision-critical environments, where both high predictive performance and interpretability are essential, The lack of transparency in GNN decisions significantly limits their applicability. The existing models usually focus on instance-level or model-level explainability to explain GNNs. Instance-level explanation methods focus on how to explain a model's prediction for a given graph instance. Model-level explanation methods focus more on understanding the general behavior of the model not specific to any particular graph instance.

4.1. Instance-level explanations

GNNExplainer [61] is the first model-agnostic interpretable approach to a GNN-based model on any graph-based task. It optimizes soft masks for edges and node features to maximize the mutual information between the predictions and new predictions from possible subgraph structures. It returns instance-based explanation with a subgraph structure and a subset of node features that are crucial in GNN's predictions.

Grad-CAM [62,63] uses the gradients of any target concept, such as the motif in a graph flowing into the final convolutional layer, to produce a coarse localization map highlighting the critical regions in the graph for predicting the concept. ACGAN-GNNEplainer [64] uses the auxiliary classifier GAN [65] as its backbone to generate explanations for GNNs. Through an iterative interplay learning process between the generator and the discriminator, the generator can ultimately produce explanations that reveal underlying patterns of graphs on a goal scale.

MatchExplainer is a non-parametric subgraph matching framework aiming to explore the explanatory subgraphs [66]. It explores the most crucial joint substructure by minimizing their node corresponding-based distance in the high-dimensional feature space. Then the target graph is compared with other counterpart graphs in the reference set to seek potential explanatory subgraphs. Lastly, it proposes to optimize the final candidate explanations by maximizing the difference in the prediction after the explanatory subgraph is removed from the original graph.

Shapley Explainer [67] is an interpretation method that combines Shapley values with a soft discrete mask matrix. Shapley values are set as the scoring function to fairly generate node contributions for each input of GNN. To address the excessive computation cost of Shapley values, They propose a trainable soft discrete mask matrix to select important nodes subset.

4.2. Model-level explanation

XGNN [68] frames the Global Explanation problem for GNNs as a form of input optimization using policy gradient to generate synthetic prototypical graphs for each class with some prior knowledge. Gem [69] formulates the problem of providing explanations for the decisions of GNNs as a causal learning task. The notion of Granger causality is then extended to characterize the explanation of an instance by its local subgraphs and provide an approximate computation strategy to deal with graph data with interdependency. No requirements on GNN structure or prior knowledge of learning tasks are needed. Gem is fast to explain once trained. GLGExplainer [70] provides a Global Explanation in terms of logic formulas, extracted by combining in a fully differentiable manner graphical concepts derived from local explanations. It is also faithful to the data domain. Therefore the logic formulas derived from local explanations are intrinsically part of the input domain without requiring any prior knowledge.

GNN Interpreter [71] learns a probabilistic generative graph distribution that can identify the most discriminative graph pattern relevant to the prediction task of GNN. This is achieved through optimizing a unique objective function considering the similarity between the explanation graph embedding and the average embedding from the target class. The work [72] focuses on the theoretical analysis of the reliability of state-of-the-art GNN explanation methods. It theoretically analyzes the behavior of various state-of-the-art GNN explanation methods with respect to several desirable properties, e.g. faithfulness, stability, and fairness preservation, and establishes upper bounds on the violation of these properties.

5. GNN theory

5.1. Graph augmentation

GNNs have been criticized for their limited ability to make full use of unlabeled data [28,73], which is vital in graph-based semi-supervised learning (SSL) method. Previous findings that used pseudo labels [74] to solve this limitation still suffered from poor calibration.

Consistency and diversity metrics have been proposed to evaluate the quality of graph augmentations in the aspects of augmentation correctness and generalization [75]. The work [75] points out that label propagation utilizes the prior knowledge of graphs to improve consistency and consistency regularization adopts variable augmentations to promote diversity. Feng et al. [76] aimed to reactivate redundant channels to enlarge the representation capacity of GNNs for effective graph learning. AKE-GNN performs the Adaptive Knowledge Exchange strategy among multiple graph views generated by graph augmentations. AKE-GNN trains multiple GNNs to obtain informative channels and iteratively exchanges redundant channels with informative channels of another GNN in a layer-wise manner.

5.2. Expressivity

One of the most significant concerns of GNNs is their limitation in expressive power. The Weisfeiler-Lehman graph isomorphism test by [77] is a powerful test known to distinguish a broad class of graphs. Recent studies [78,79] introduce the Weisfeiler-Lehman (WL) hierarchy as a measure of expressive power for GNNs. It has become standard to characterize GNN architectures in terms of the separation power of graph algorithms such as color refinement (CR) and k-dimensional Weisfeiler-Leman tests.

Zhang et al. [80] discover that many GNN architectures including 1-WL, Substructure Counting WL [81] and GNN-AK [82] cannot distinguish biconnectivity problems. Therefore, they propose a Generalized Distance Weisfeiler-Lehman (GD-WL) which can be provably expressive for all biconnectivity metrics. Geerts et al. [83] provid a tensor language-based technique to analyze the separation power of general GNNs. Omri et al. [84] point out several limitations in WL hierarchy which includes a complex definition that lacks direct guidance for model improvement and that WL hierarchy is too coarse to study GNNs. They thus introduce an alternative expressive power hierarchy called polynomial expressiveness based on the ability of GNNs to calculate equivariant polynomials of a certain degree.

5.3. Over-smoothing

Li et al. [85] show that the neighborhood aggregation of GCNs is essentially a form of Laplacian smoothing and too many convolutional layers may lead to over-smoothing. A canonical metric for over-smoothing is average similarity [86,87]. The tendency of similarity converging to 1 indicates that representations shrink to a single point (complete collapse) [88].

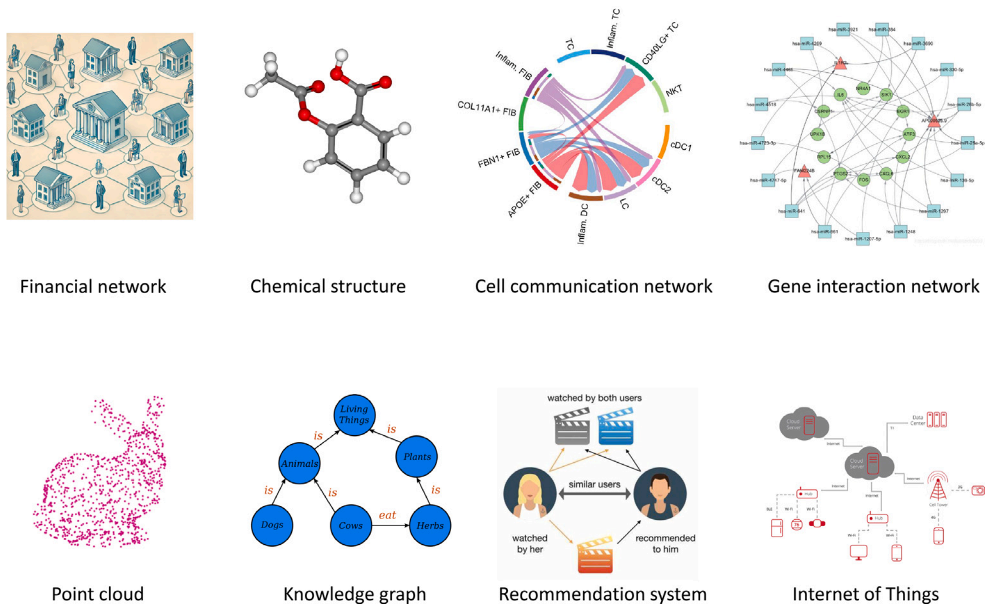

Fig. 4. Examples of applications in GNN.

MatchDrop technique [66] is introduced to solve false positive sampling problems in present graph sampling or node-dropping mechanisms. The work [66] first digs out the explanatory subgraph through MatchExplainer and keeps this part unchanged. Then the graph sampling or node dropping is implemented solely on the remaining less informative part. This ensures that the core fraction of the input graph that reveals the label information is not affected, thus greatly mitigating the false positive sampling issue.

For the normalization layer, Guo et al. [88] introduce ContraNorm to implicitly shatter representations in the embedding space which can lead to a more uniform distribution and a slighter dimensional collapse. PairNorm [89] is based on analysis of the graph convolution operator. It ensures that the total pairwise feature distances remain constant across layers, which in turn leads distant pairs to have less similar features, thus preventing feature mixing across clusters.

Decoupled GCN has shown good performance when dealing with over-smoothing like APPNP [90], and SGC [91]. APPNP [90] is a GCN model with an improved propagation scheme based on personalized PageRank. It separates a neural network from the propagation scheme and uses PageRank to balance the needs of preserving locality and leveraging the information from a large neighborhood without leading to over-smoothing. Simple Graph Convolution (SGC) [91] shows that the complexity of GCNs may not always be necessary for effective learning from graph data by using only linearized GCN transformation to maintain comparable performance to GCN. SGC removes nonlinearities in GCN layer-wise propagation and collapses multiple weight matrices between two consecutive layers into a single linear transformation.

Dong et al. [92] point out that the working mechanism of decoupled GCN remains unclear. They explore the decoupled GCN for semi-supervised node classification from label propagation perspective and reveal that decoupled GCN can automatically assign structure-aware and model-aware weights to the pseudo-label data, thus resulting in a relatively robust GCN to structure noise and over-smoothing.

6. Applications of GNNs

GNNs have gained significant applications in recent years as a powerful tool for analyzing interconnected data represented in graph structures. Unlike standard neural networks that operate on arrays, GNNs are specifically designed to work on graphs, making them suitable for a wide range of applications where data exhibits relational structures [26,93–95]. Here we outline several important applications summary in Table 3 and different application scenarios in Fig. 4.

6.1. Finance

Financial systems, characterized by their numerous components, complex relationships, and frequent updates, well fall into the application scope of graph representation methods that depict transaction networks, user-item review relationships, and the connections among stocks. Converting financial tasks into node classification tasks that GNNs can effectively handle, such as stock prediction, demonstrates the potential of GNN methods in solving practical problems. However, the multifaceted nature of financial data sources, complexity of graph structures, and unique temporal variability and heterogeneity of financial systems pose challenges to the practical application of GNNs.

Facing data quantity-imbalance and topology-imbalance issues in finance, Liu et al. [96] designed a multi-hierarchy label-aware neighbor sampling strategy based on random walk with restart to ensure that the minority nodes have enough neighbors of the same class and a class-balanced loss to achieve an optimal trade-off between majority and minority classes in terms of bankruptcy prediction accuracy.

Song et al. [97] proposed two individual fairness properties on temporal graphs and designed a temporally fair GNN framework on dynamic transaction networks accordingly. They aim to ensure fair financial service for all clients.

In financial fraud detection, different neighbor node selection strategies are proposed to improve the detection outcome. FA-GNN [98] uses three priority policy neighbor filtering strategies to find the vital neighbor nodes. RioGNN [99] adopts an enhanced relationship-aware neighbor selection mechanism to identify the most similar neighbors from multiple relationships via label-aware neural similarity measure. TA-Struc2Vec [100] utilizes both structural homogeneity and transaction financial homogeneity to identify long-distance similar nodes.

6.2. Biology

In the field of bioinformatics, GNNs are particularly well-suited for analyzing biological data due to the inherent complexity and interconnected nature of biological systems. Biological data often consists of various types of entities and their intricate relationships, such as proteins and their interactions [34], genetic networks, and cellular pathways. GNNs excel in capturing these relationships by representing them as graphs, where nodes can represent entities like genes or proteins, and edges represent their interactions or relationships. By

Table 3
Applications of graph neural networks.

Field	Contents	Reference
Finance	Service fairness	Liu et al. [96], Song et al. [97]
	Fraud detection	Liu et al. [98], Li et al. [100], Peng et al. [99]
Biology	Medical disease analysis	Abdous et al. [101], Zhao et al. [102], Biswas et al. [103]
	Protein structure	Wang et al. [104], Mastropietro et al. [105]
	Single-cell data	Wen et al. [3]
Knowledge graph	Knowledge embedding	Li et al. [2], Schlichtkrull et al. [107], Xu et al. [108], Pflueger et al. [109], Cucala et al. [110], Meng et al. [111]
Computer vision	Point cloud	Wang et al. [112], Qin et al. [113]
	Interaction prediction	Rowe et al. [114]
	Image segmentation	Aflalo et al. [115], Yang et al. [116]
Recommendation systems	Session-based recommendation	Zhang et al. [117], Liang et al. [118], Dai et al. [119]
	Social recommendation	Wang et al. [120], Liu et al. [121]
	Sequential recommendation	Hao et al. [122]
Internet of Thing (IoT)	Industrial network	Liu et al. [123], Bi et al. [124], Bi et al. [125], Liu et al. [126]
	Social network	Wang et al. [1]
Knowledge distillation	Data-free KD	Zhuang et al. [127]
	Topology KD	Zhou et al. [128], Xia0 et al. [129], Feng et al. [130], Yang et al. [131]
	GNN-to-MLP KD	Wu et al. [132]

using GNNs, we can learn more about the properties and dynamics of biological networks, leading to deeper insights and more accurate predictions.

On histopathology images, Abdous et al. [101] implemented GNNExplainer on histopathology images patterns discovering with sample Kolmogorov–Smirnov test score as objective function. The EAGMN [102] utilizes an edge attention mechanism to efficiently capture the significance of different edge connections and feature representation within a graph to improve ICA image matching accuracy for coronary artery semantic labeling. Biswas et al. [103] propose a new framework to perform link analysis tasks on a comorbid disease-centric KG. They compare four different heterogeneous GNN models on comorbid disease-centric KG and predict novel biomarkers and therapeutics for complex comorbid diseases.

ProS-GNN [104] is a gated GNN based approach to predict changes in protein stability upon mutation. Wang et al. [104] focus on protein mutant structure and encoded the structural information and element compositions as node features for gated GNN model. In addition, they incorporate the coordinates of the raw atoms to provide spatial insights into the chemical systems.

In drug design, GNN is a popular machine learning tool used to predict the potency of compounds. Different GNNs [105] have been explored in protein-ligand affinity prediction. The studies [105] have performed affinity predictions with six GNN architectures on community-standard datasets and rationalized the predictions using EdgeSHAPer [106] and GNNExplainer [61]. They suggest that ligand memory effect plays a dominant role in predicting protein-ligand affinity. However, these models increasingly prioritize interaction information in predicting high affinity situations.

Multimodal single-cell technologies have been used to provide deeper insights into cellular states and dynamics [3]. It is rather challenging to incorporate the vast number of single-modality datasets into the downstream analyses. [3] focuses on three key tasks of multimodal single-cell data analyses and presents a general GNN-based framework scMoGNN to capture the high-order structural information between cells and modalities.

6.3. Knowledge graph

Knowledge graphs (KGs) are powerful tools for representing real-world entities and their interrelations, widely used in applications like question answering, information retrieval, and knowledge graph reasoning [133]. There are two main approaches to KG processing: Conventional Knowledge Graph Embeddings (KGEs) and Graph Neural Network (GNN)-based methods. Conventional KGEs focus on validating triples using geometric assumptions in vector space and are exemplified by methods like AnKGE [134] and KGE-HAKE [135]. On the other hand, more recent work focuses on GNN-based methods, such as the pioneering SE-GNN [2].

The work [107] makes the first attempt to apply GCN framework to modeling relational data by presenting R-GCN. It introduces relation-specifc transformations with sparsity constraints when aggregating multigraphs. SE-GNN [2] uses three levels of Semantic Evidence (SE) to understand the extrapolation ability of KGE. In SE-GNN, each level of SE is modeled explicitly by the corresponding neighbor pattern and merged sufficiently by the multi-layer aggregation, which contributes to obtaining more extrapolative knowledge representation. MA-GNN [108] is a double-branch multi-attention-based GNN that enhances entity representations using both global and local structural information from KGC. It uses a snowball local attention mechanism to capture entity similarity among 2-hop neighborhood and a Transformer-based self-attention mechanism for long-range entity inter-dependency representations from the global graph structure. Meng et al. [111] uniquely leverages high-frequency relations for enhanced relation reasoning. They select relevant auxiliary relations based on semantic similarity and integrate summation and learnable fusion strategies to capture co-occurrence patterns of auxiliary relations.

GNNQ [109] is a novel neuro-symbolic approach to query answering over incomplete KGs applicable in the inductive setting. It uses symbolic rules to augment the input KG with facts representing subgraphs and encodes the augmented KG into a hypergraph. Then it processes the hypergraph using HRGCN to produce the predicted query answers. MGNN [110] is a GNN-based transformation of graph data whose predictions can be explained symbolically as logical in inferences in Datalog. MGNN keeps transformation monotone under homomorphism property. A graph counterfactual augmentation method is proposed in [136] for knowledge tracing.

6.4. Computer vision

GNN applications for computer vision include generating scene graphs, point cloud classification, and action identification, and so on.

Wang et al. [112] proposes a Progressive Quadric Graph Convolutional Network (PQ-GCN), and the authors have designed a simple and fast method for 3D human mesh recovery from a single image in the wild. A robust global descriptor is designed for the classification task of complex real-world point clouds [113]. It utilizes non-local graph attention network to integrate both geometric structure and point-point relation information from the point cloud that can preserve valuable low-frequency features for noisy point clouds and guarantee invariance to rotational transformations.

FJMP [114] models the future interaction scene of road agents and generates agents' interaction directed acyclic graph (DAG) based on explicit interactions between agents. The work [114] decomposes the joint prediction of trajectories into a sequence of marginal and conditional predictions based on partial ordering of the DAG.

Aflalo et al. [115] transform images into graph representations through patch-wise feature similarities on extracted deep features and turn image segmentation problem into node clustering problem. The work optimizes the correlation clustering objective for deep features clustering and achieving k-less clustering. Another work [116] approaches semantic segmentation of floorplans through the dual aspects of boundary line classification and region classification based on graph attention network.

6.5. Recommendation systems

With the rapid growth of the amount of information on the Internet, Recommendation Systems become significant to help users select interesting information within a variety of Web application domains. The problem of information overload makes it increasingly difficult for users to mine useful information in the age of information explosion, and recommendation systems can assist users in discovering their truly needed information. Session-based recommendation predicts a user's subsequent interactions based on a session, which is a set of items in a sequence. According to [137], by combining GNN and session, higher-order features can be learned while weakening the influence of strict-order relationships. According to [138], GNNs are becoming increasingly popular in session-based recommendation and have achieved state-of-the-art performance.

To better obtain the representation of static information from one session, DSGNN [117] is introduced by combining the representation of dynamic and static information extracted from digraph and undigraph with a lightweight gate mechanism to better combine a user's recent and relatively long-term intentions. To consider the influences of different interactive behaviors and the noise in interactive sequences, Liang et al. [118] put forward a behavior-aware GNN (BA-GNN) for session-based recommendation. A sparse self-attention module is used in BA-GNN for interactive behavior sequences denoise. Then different behavior sequences are aggregated with a gating mechanism to obtain the session representations. Dai et al. [119] point out that the user's multiple interests may be distributed in different positions of the session and current Session-based recommendation GNNs would obtain one-sided long-term interest only. To address this issue, they propose a denoising autoencoder integrated with Self-Supervised Learning (SSL) in GNN (DAS-GNN). DAS-GNN with a query extraction module based on their denoising autoencoder can mine multiple user interests and assist long-term interest to express user needs more comprehensively.

Liu et al. [121] point out that most of the current recommendation methods are based on a single recommendation domain (e.g., session-based recommendation models or social recommendation models). They proposed GNNRec combined session-based recommendation system with social recommendation system. Thus they can resolve the shortsighted issue with RNN on session-based information to produce more accurate recommendation prediction. GNNRec uses gated GNN to process session sequences to learn the complex transition relationships of items and GAT [139] to aggregate user social information.

Besides traditional structural information and topology of user social relationships, CENTRIC [120] emphasized more on context information during user interactions to gain more accurate user interest features. It uses multi-channel aggregation to enhance user, item, and context features through four collaborative channels and considered implicit feedback in user and item features to alleviate data sparsity. It then applies tensor factorization to fuse user-item-context features for a more nuanced and comprehensive representation of user interests.

Hao et al. [122] introduced IMGC-GNN to better separate attribute factors revealed by independent individuals and interaction factors contained in the collaborative signals representation learning process. IMGC-GNN is a three-layer heterogeneous graph using historical interaction information. With implicit relationships, they construct homogeneous graphs with users, contexts, or apps as nodes for attribute representation learning. By using node similarity, they constructs interaction graphs with user-context-app interactions as nodes to mine interaction representations and makes preference predictions based on representations of attributes and interaction.

6.6. Internet of Things (IOT)

IoT is a rapidly growing field that significantly impacts various aspects of daily life, such as healthcare, environment, transportation, manufacturing [93], and supply chains. The complexity of GNN allows for a more effective representation of the complex interactions and dependencies between IoT devices and their data collected over time.

A knowledge graph-based system is employed for cognitive manufacturing [123], offering a hierarchical integration of IoT data. The system enables machine perception and cognitive responses through its perception and cognition modules. The structure also processes and automatically integrates multi-source data from personalized products, while decision-making and reasoning are aided by the graph embedding method.

The minority-weighted Graph Neural Network (mGNN) [1] is proposed to focus on the imbalanced problem of reality social networks in the Internet of People (IoP). mGNN uses weighted aggregation to enhance the impact of minority nodes and applies cost-sensitive learning with a higher misclassification cost to minority nodes on edge predictor during oversampling operation. Additionally, the Gumbel distribution activation function is used to overcome the problem of underestimating the probability of minority nodes in ReLU.

Bi et al. [124] focus on the representation learning of high-dimensional and incomplete industrial data. They introduce proximal regularization into the alternating-direction-method-of-multipliers model that inherits model's fast convergence and high computational efficiency while reducing optimization fluctuations and enhancing representation learning accuracy. Nonnegativity constraints are introduced into the autoencoder and data density-oriented mechanism is integrated to reduce reconstruction errors and enhance latent feature extraction efficiency [125].

For large-scale undirected industrial networks, Liu et al. [126] introduced the Symmetry and Graph Bi-regularized Non-negative Matrix Factorization method aimed at community detection. This model integrates symmetry and graph geometry regularization, which enhances the latent factor space and significantly improves the representation capability of the method, making it particularly effective for large-scale undirected networks.

6.7. Knowledge distillation

Knowledge Distillation (KD) has demonstrated its effectiveness to boost the performance of GNNs, where its goal is to distill knowledge from a deeper teacher GNN into a shallower student GNN. This has enabled remarkable progress in graph model compression and knowledge transfer.

DFAD-GNN [127] is an end-to-end framework for data-free adversarial knowledge distillation on graph-structured data. DFAD-GNN allows a student model to learn effectively from a pre-trained teacher model based on synthetic data generated through an adversarial process, ensuring knowledge transfer in a privacy-preserving manner. This adversarial framework is suitable for situations where access to the original data is restricted or impossible.

Facing with over-parametrized and over-smoothing issues, we often arrive at a less satisfactory teacher GNN model. This problem leads to

Table 4
Graph datasets.

Field	Datasets	Description
Social networks	Reddit [140]	A dataset of posts and comments from the website of Reddit
	Twitch Egos [141]	A collection of ego-networks from Twitch users
	GitHub Repos [141]	Collections of files and folders associated with a software project
Bio-chemical graphs	PPI [142]	A collection of protein-protein interaction data, which is essential for understanding the biological processes within organisms.
	OGBL-ppa [143]	Protein-Protein Association Network designed for link property prediction tasks
	OGBL-ddi [143]	Drug-drug interaction network for predicting the effects of drugs when taken together
	MoleculeNet [144]	A large-scale benchmark for molecular machine learning designed to assess the performance of models on various molecular properties
Knowledge graphs	OGBL-biokg [143]	A biomedical knowledge graph constructed from various repositories, encompassing entities like diseases, proteins, drugs, side effects, and protein functions, with directed relations
	OGBL-wikikg2 [143]	A comprehensive knowledge graph extracted from Wikidata, featuring a diverse set of entities and relations that capture various factual associations between them, with over 2.5 million entities and 17 million edges.
	WN18RR [145]	A subset of WordNet that focuses on 18 relations and contains 18,000 entities, used for the task of link prediction in knowledge graphs
	WN18 [146]	A subset of WordNet for the task of knowledge graph completion and link prediction in the field of knowledge representation learning
Citation networks	DBLP [147]	A comprehensive computer science bibliography that includes metadata for millions of publications, authors, and venues
	Patent [148]	A vast collection of patent documents, images, and metadata that provide a rich source for research in technical drawing understanding, patent information retrieval, and patent representation learning
	MAG [149]	A large-scale heterogeneous academic graph derived from the Microsoft Academic Graph, consisting of 121 million academic papers, 122 million authors, and 26 thousand institutes, with 1.3 billion citation links, aimed at predicting the primary subject areas of arXiv papers using their titles and abstracts, and is designed to support research in knowledge graph completion and academic trend analysis

invalid knowledge transfer in practical applications. A Free-direction Knowledge Distillation (FreeKD) framework [130] is proposed with the help of Reinforcement learning to avoid the use of deep well-optimized teacher GNN. FreeKD can dynamically manage the directions of knowledge transfer among both node-level and structure-level via a hierarchical reinforcement learning algorithm.

HKD [128] distills the holistic knowledge that is the integration of both individual knowledge and relational one for student network learning. HKD uses topology GCN on an attributed context graph to extract the holistic knowledge and aims at maximizing the mutual information on graph-based representation with InfoNCE estimator [150].

Topology of graph is an essential part of the information for GNN input. Neural Heat Kernel (NHK) [131] tries to encode geometric properties of the manifold behind GNNs into a series of matrix representations between layers. Through geometric knowledge distillation, they achieve the compression of graph topology information into the model itself and realize knowledge transfer between different GNNs by aligning the neural heat kernels of both teacher and student GNN models.

ARGD [129] aims to eliminate backdoor triggers by purifying the malicious impacts of backdoored models. Inspired by NAD [151], ARGD combines attention relation graph with knowledge distillation to reflect discriminative regions in the model's topology. In addition, ARGD takes the correlations among attention features of different orders into consideration instead of the same order in NAD.

With MLPs being the primary workhorse for practical industrial applications, directly distilling knowledge from a well-designed teacher GNN to a student MLP is proposed to reduce the gaps between GNNs and MLPs. The work [132] identifies a potential information drowning problem in GNN-to-MLP distillation and proposes an efficient Full-Frequency GNN-to-MLP (FF-G2M) distillation framework. FF-G2M can help extract both low-frequency and high-frequency knowledge from GNNs and inject it into MLPs.

7. Evaluation and dataset

In real-world applications, there is increasing focus on experimental reproducibility and replicability. While numerous GNN methods have emerged over the years, it remains crucial to determine the extent to which these models are effective. To address these fundamental questions, studies and open datasets that enable fair comparison and evaluation of GNNs are essential. We have summarize some of the commonly used graph datasets in Table 4.

To facilitate a comprehensive comparison of diverse algorithms, Xiao et al. [152] provided an overview of four commonly used evaluation metrics for classification tasks and assessed the performance of several popular GNN models based on these criteria. You et al. [153] concentrated on the architectural design aspects of GNN models, offering insights into potential pitfalls. Furthermore, Thompson et al. [154] conducted an extensive review of GNN model explainability, evaluating it from multiple perspectives, including fidelity, sparsity, stability, and fairness. The open-source code links related to the GNN methods have been summarized in Table 5.

8. Challenges in graph neural networks

Although GNNs have succeeded in many tasks and applications, they cannot provide acceptable answers for graph structures in some other situations. Several of the following challenges need to be investigated further.

8.1. Dynamic graph neural networks

Traditional GNNs assume static graphs, where the structure remains constant. In dynamic scenarios, edges and nodes can be added or removed over time. Dynamic graph networks become increasingly important for addressing real-world problems where relationships are not static in numerous cases. Dynamic GNNs find applications in various domains, such as social network analysis, traffic prediction, recommendation systems, and biological systems where the relationships among entities tend to evolve sometimes rapidly. Modeling dynamic graphs introduces challenges such as preserving information across time steps, handling irregularly sampled data, and efficiently updating representations.

Table 5
Source code of the models mentioned in the survey.

Model	Link
TGLFA(2023)	https://github.com/Oak-B/TGLFA
NODE-SELECT (2022)	https://github.com/superlouis/NODE-SELECT
GRAND (2020)	https://github.com/THUDM/GRAND
GRAND+ (2022)	https://github.com/THUDM/GRAND-plus
LazyGNN (2023)	https://github.com/RXPHD/Lazy_GNN
NWR-GAE (2022)	https://github.com/mtang724/NWR-GAE
GraphMAE (2022)	https://github.com/THUDM/GraphMAE
Bi-GAE (2023)	https://github.com/citsjtu2020/BiGAE
GAT (2018)	https://github.com/PetarV-/GAT
MAGNA (2021)	https://github.com/xjtuwgt/GNN-MAGNA
HAGNN (2024)	https://github.com/PasaLab/HAGNN
MGAT (2023)	https://github.com/zltao/MGAT/tree/master
MagNet (2021)	https://github.com/matthew-hirn/magnet
HGSL (2021)	https://github.com/AndyJZhao/HGSL
SeHGNN (2023)	https://github.com/ict-gimlab/sehgnn
GReTo (2023)	https://github.com/zzyy0929/ICLR23-GReTo
TGC (2024)	https://github.com/MGitHubL/Deep-Temporal-Graph-Clustering
NAT (2022)	https://github.com/Graph-COM/Neighborhood-Aware-Temporal-Network
TMac (2023)	https://github.com/MGitHubL/TMac
DGIB (2024)	https://github.com/haonan-yuan/DGIB
HGNN+ (2022)	https://github.com/iMoonLab/DeepHypergraph
HCoN (2023)	https://github.com/wuhanrui/HCoN
GReTo (2023)	https://github.com/zzyy0929/ICLR23-GReTo
TGC (2024)	https://github.com/MGitHubL/Deep-Temporal-Graph-Clustering
NAT (2022)	https://github.com/Graph-COM/Neighborhood-Aware-Temporal-Network
TMac (2023)	https://github.com/MGitHubL/TMac
DGIB (2024)	https://github.com/haonan-yuan/DGIB
HGNN+ (2022)	https://github.com/iMoonLab/DeepHypergraph
HCoN (2023)	https://github.com/wuhanrui/HCoN
GReTo (2023)	https://github.com/zzyy0929/ICLR23-GReTo
TGC (2024)	https://github.com/MGitHubL/Deep-Temporal-Graph-Clustering
NAT (2022)	https://github.com/Graph-COM/Neighborhood-Aware-Temporal-Network
TMac (2023)	https://github.com/MGitHubL/TMac
DGIB (2024)	https://github.com/haonan-yuan/DGIB
HGNN+ (2022)	https://github.com/iMoonLab/DeepHypergraph
HCoN (2023)	https://github.com/wuhanrui/HCoN

– *Temporal Dependency Modeling*: DyGNNs require sophisticated mechanisms to capture and model the temporal evolution of graphs. This necessitates an understanding of not only the current graph state but also the ability to anticipate or adapt to future structural changes, integrating time series analysis and forecasting techniques.
– *Information Propagation Mechanisms*: The transmission of information across time steps in dynamic graphs is a central challenge. Designing models that can effectively propagate and update information while maintaining coherence and accuracy is paramount.
– *Irregular Data Sampling*: Real-world graph data is often irregularly sampled, with varying update frequencies and intervals. DyGNNs must be robust against such irregularities, ensuring model generalizability and resilience.
– *Efficient Representation Updating*: As the graph structure evolves, the representations of nodes and edges must be updated accordingly. Achieving efficient representation updates without compromising quality is a technical challenge that requires innovative solutions.
– *Multi-scale and Multi-dimensional Modeling*: Dynamic graphs may exhibit behaviors at different temporal scales and dimensions. Designing models that can handle multi-scale and multi-dimensional information to capture a comprehensive view of entity relationships and changes is an important research direction.

8.2. Scalability of GNN methods

Many real-world graphs are large and complex. Scaling GNN methods to handle large graphs while maintaining efficiency is a significant challenge. Many real-world graphs are sparse, meaning that most nodes are not directly connected. Exploring model parallelism, where different parts of the model are processed on different devices, can be challenging in the context of large-scale GNNs. Hence, efficiently handling sparse graphs without unnecessary computations, and partitioning and distributing the model components to multiple parallel processors is essential for raising a GNN method's scalability.

– *Computational Complexity*: Traditional GNNs often have a quadratic complexity with respect to the number of nodes due to the message passing mechanism, which can become computationally expensive for large graphs. Methods that reduce complexity, such as using sparse matrix operations or approximate computing techniques, are essential for improving scalability.
– *Parallelization*: Leveraging parallel computing resources can significantly enhance the scalability of GNNs. However, parallelizing GNNs can be challenging due to dependencies between nodes introduced by the message passing process.
– *Subgraph-based Methods*: Processing large graphs by considering subgraphs or clusters can reduce the complexity and improve scalability. Techniques such as graph coarsening or community detection can be used to focus computation on relevant parts of the graph.
– *Distributed GNNs*: Distributed computing frameworks can be employed to scale GNNs horizontally across multiple machines or processors, allowing for the processing of larger graphs than would be possible on a single machine.
– *Sampling Techniques*: Node or edge sampling techniques can be used to create smaller, representative subgraphs for training, which can significantly reduce the computational and memory footprint of GNNs.
– *Adaptive Computation*: Adaptive methods that allocate more computation to parts of the graph that are more complex or informative can improve both the efficiency and scalability of GNNs.

– *Hybrid Models*: Combining GNNs with other machine learning models, such as attention mechanisms or recurrent neural networks, can provide more flexibility and scalability in handling large and complex graphs.

8.3. Transfer learning of GNNs

Extending GNNs to perform well in transfer learning tasks, where knowledge gained from one graph domain can be applied to another, is a challenging task. Transfer learning in GNNs involves leveraging knowledge learned from one graph domain to improve performance on a different but related graph domain. While transfer learning has shown success in various machine learning domains, there are specific challenges when applying it to GNNs, including graph structure variability, heterogeneous graphs, task misalignment in graphs, and so on. Ensuring that the learned representations generalize well across varying graph topologies is a key challenge. For instance, aligning nodes or edges across different graphs is a non-trivial task. Ensuring that nodes with similar semantics in different graphs are aligned correctly is crucial for successful transfer learning.

– *Graph Structure Variability*: Graphs across different domains may exhibit significant topological disparities, making it difficult for knowledge learned in one domain to be directly applicable to another. The variability in graph structure demands that GNNs adapt to diverse graph patterns.
– *Heterogeneous Graphs*: Real-world graphs are often heterogeneous, containing a variety of node and edge types. The complexity of heterogeneous graphs introduces additional challenges for transfer learning, as different types of nodes and edges may require distinct handling.
– *Task Misalignment*: The tasks in the source and target domains of transfer learning may not be perfectly aligned, potentially rendering the transferred knowledge irrelevant or mismatched for the target task.
– *Distribution Shift*: The data distribution between the source and target domains may differ, and such distribution shifts can diminish the effectiveness of transfer learning.
– *Cross-Domain Generalizability*: GNNs require the capability to generalize across different domains, necessitating models that perform well not only in the source domain but also exhibit effective transfer in the target domain.

9. Conclusion

This article provides an overview of recent Graph Neural Network (GNN) methods and their industrial applications. We begin by elucidating GNN architectures, showcasing their adeptness in capturing intricate relationships and dependencies inherent in graph data. Afterward, we delve into the variants and evolution of GNNs, encompassing directed graphs, heterogeneous graphs, dynamic graphs, and hypergraphs. Following this, we explore the interpretability of GNNs and GNN theories such as graph augmentation, expressivity, and over-smoothing. Then, we elucidate specific industrial applications where GNNs demonstrate their utility, spanning diverse domains such as finance, biology, knowledge graphs, recommendation systems, the Internet of Things (IoT), and knowledge distillation. Finally, we summarize some challenges in GNN research, as well as opportunities for addressing these issues.

CRediT authorship contribution statement

Haoran Lu: Writing – original draft. **Lei Wang:** Writing – review & editing, Project administration, Conceptualization. **Xiaoliang Ma:** Investigation. **Jun Cheng:** Funding acquisition. **Mengchu Zhou:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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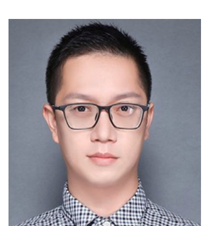

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